Chapter 10

Special Relativity

The principle of relativity

Galileo proposed the principle of relativity:

The laws of physics are the same for inertial frames that are moving with respect to each other.

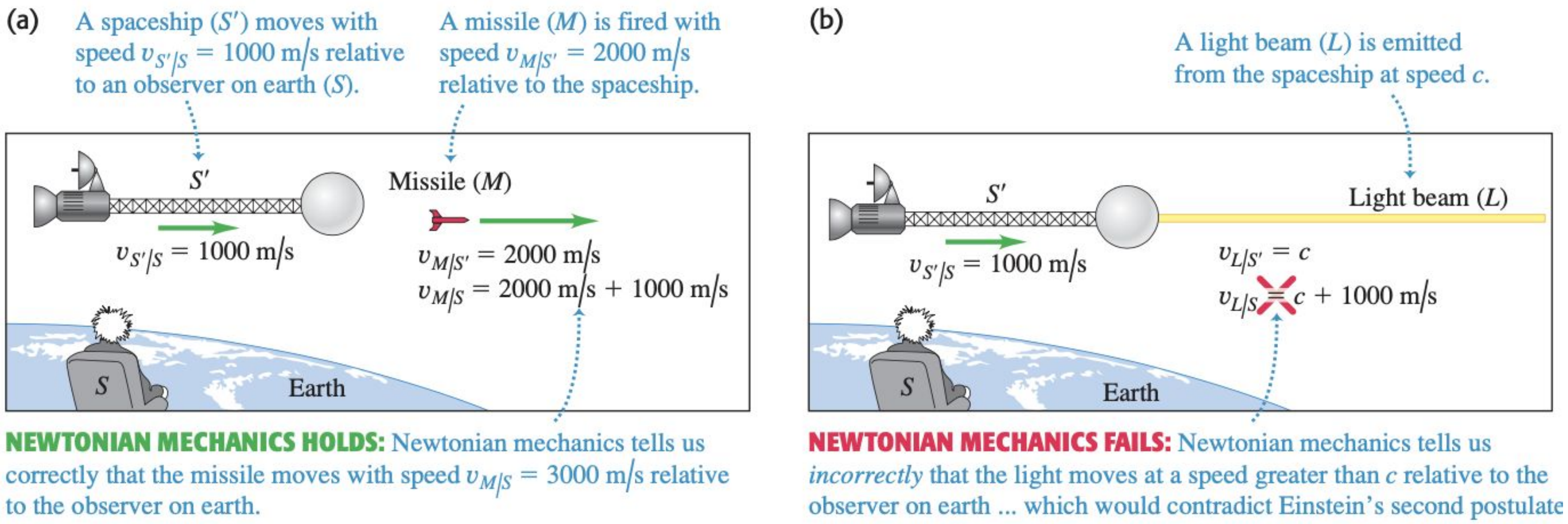
In the 19th century, Maxwell found an important law of electrodynamics:

The speed of light in vacuum equals 299 792 458 m/s

The principle of relativity

According to the principle of relativity, the law "The speed of light in vacuum equals 299 792 458 m/s" should be true in all the inertial frames. (i.e. the speed of light in vacuum is a constant)

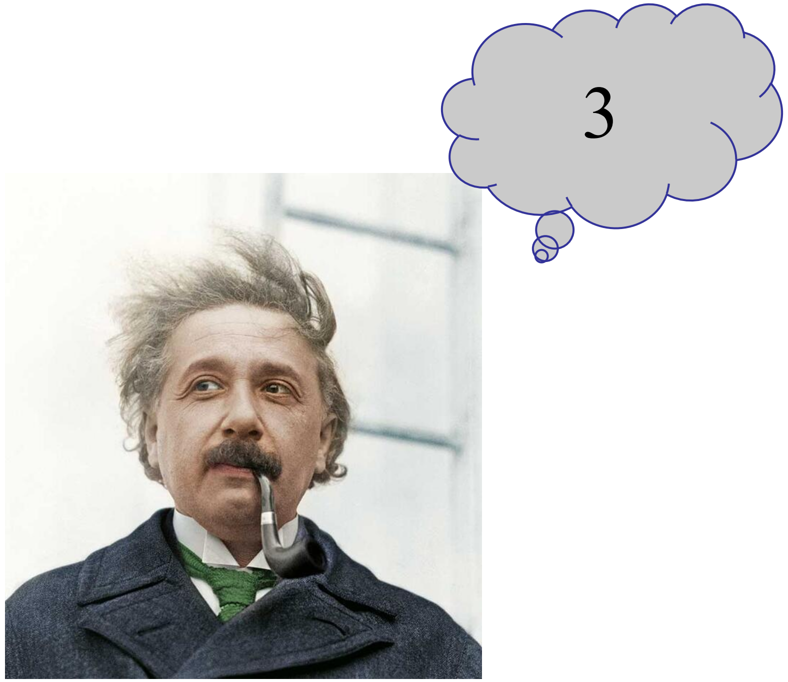
But this is in conflict with the relative velocity formula we encountered earlier.



The crisis

What shall we do?

1. Give up the principle of relativity.
2. Give up “the speed of light **in vacuum** equals 299 792 458 m/s”.
3. Give up something else.



Review Newtonian frame

Let’s first review how Newton would describe the relation between different reference frames. We’ll pretend that the world is 1D.

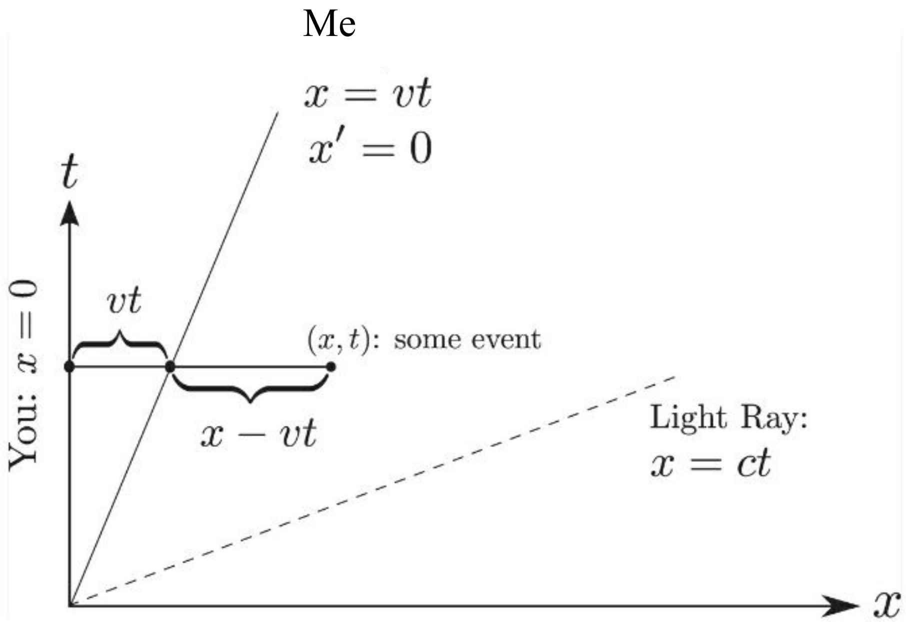


Figure 1.1: Newtonian Frames.

Review Newtonian frame

In your view (x-t frame), my motion is described by $x=vt$, or $x-vt = 0$.

In my view (x'-t' frame), my motion is described by $x'=0$.

Between the two frames, we have the relation $t'=t$, $x'=x-vt$.

It is also trivial to invert the relation $t=t'$, $x=x'+vt'$.

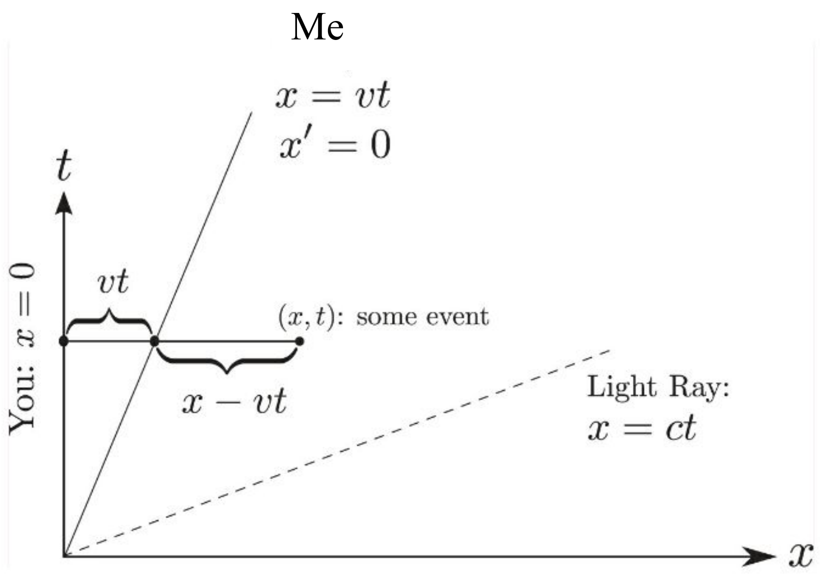


Figure 1.1: Newtonian Frames.

Review Newtonian frame

How about the light ray?

In your view, it is described by $x=ct$.

Therefore in my view, it is described by $x'+vt' = ct'$, i.e., $x' = (c-v)t'$.

What happens for a light ray moving to the left? In your view, $x = -ct$; In my view, $x' = -(c+v)t'$.

Me

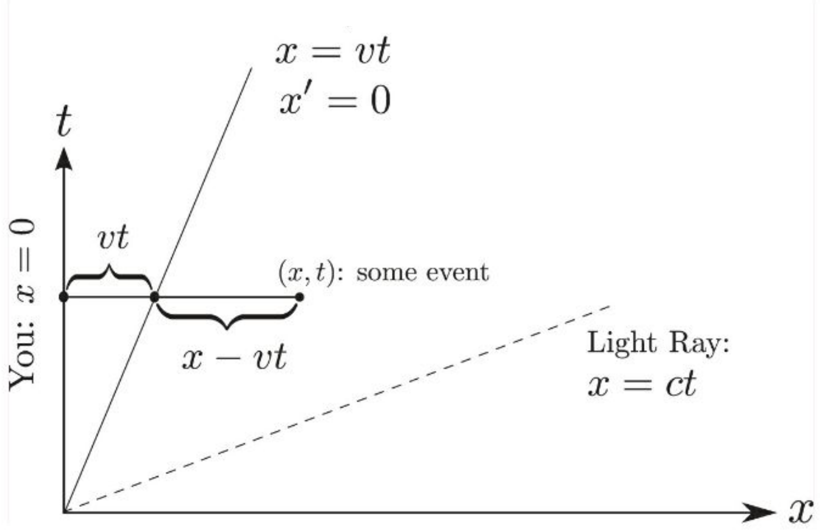


Figure 1.1: Newtonian Frames.

Review Newtonian frame

Therefore the speed of light is not a constant in Newtonian frame.

We are about to see equation $t' = t$ is not the correct relationship between moving clocks and stationary clocks. The whole idea of **simultaneity is frame-dependent**.

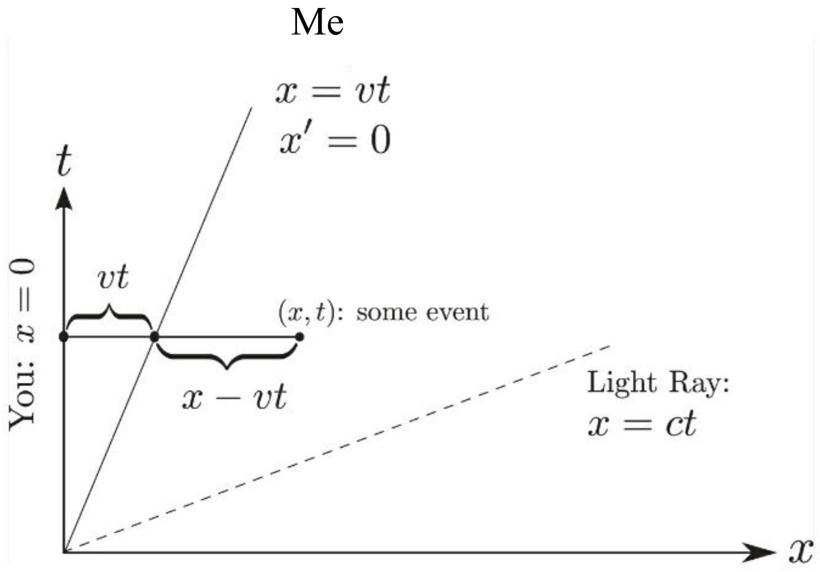


Figure 1.1: Newtonian Frames.

Change the unit

Before proceeding with the discussion, we need to change the unit of distance.

Under the SI unit, the speed of light is a huge number.

Instead, we will use “light-second” to measure the distance. One light-second equals the distance that light travels during one second.

Under the new unit system, the speed of light = 1 light-second/second.

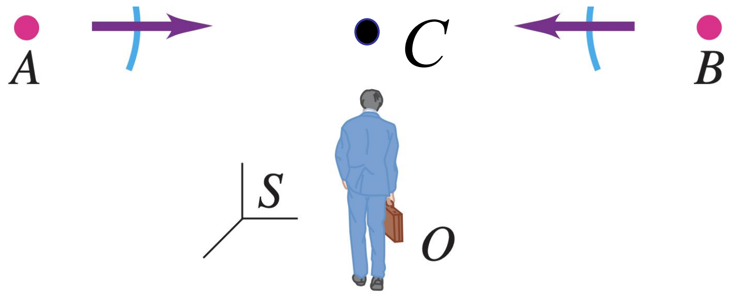
How to synchronize two clocks?

If the two clocks, say A and B, are at the same location, moving with the same velocity, it should be easy to see they are synchronized.

But even if A and B are standing still, say in your frame, but are not at the same position, checking if they are synchronized requires some thought.

Einstein's strategy was to imagine a third clock, C, located midway between A and B.

At exactly the time when the A clock reads noon it activates a flash of light toward C. Similarly when B reads noon it also sends a flash of light to C. What we mean by saying A and B are synchronized is that the two flashes will arrive at C at exactly the same time.

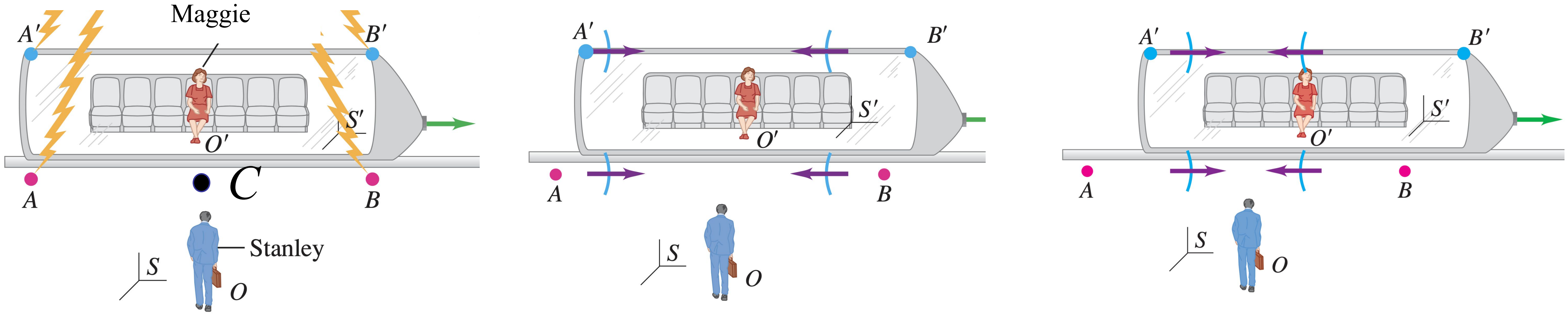


Synchronizing the Clocks

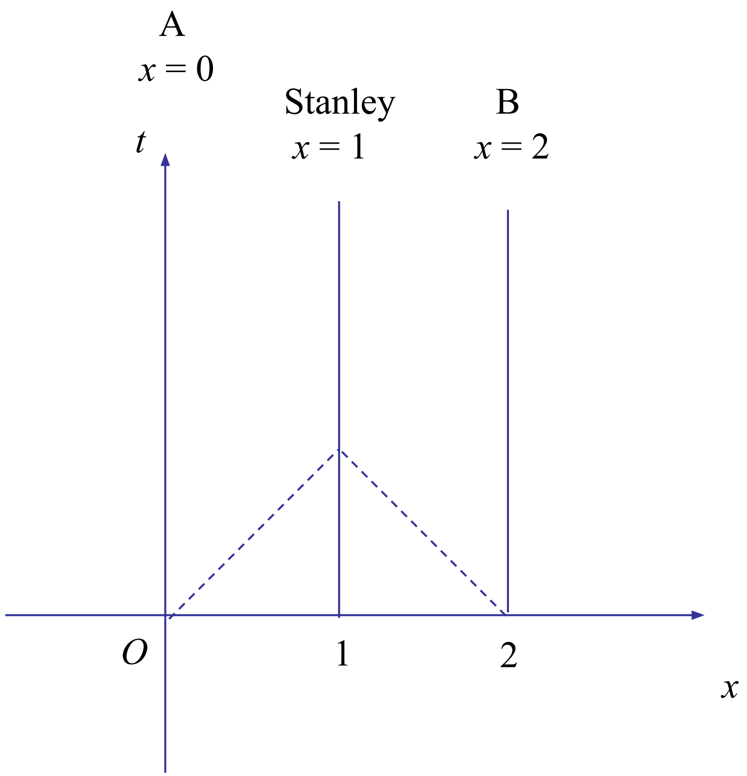
Suppose clocks A and B are synchronized in Stanley's frame. What happens to a moving observer?

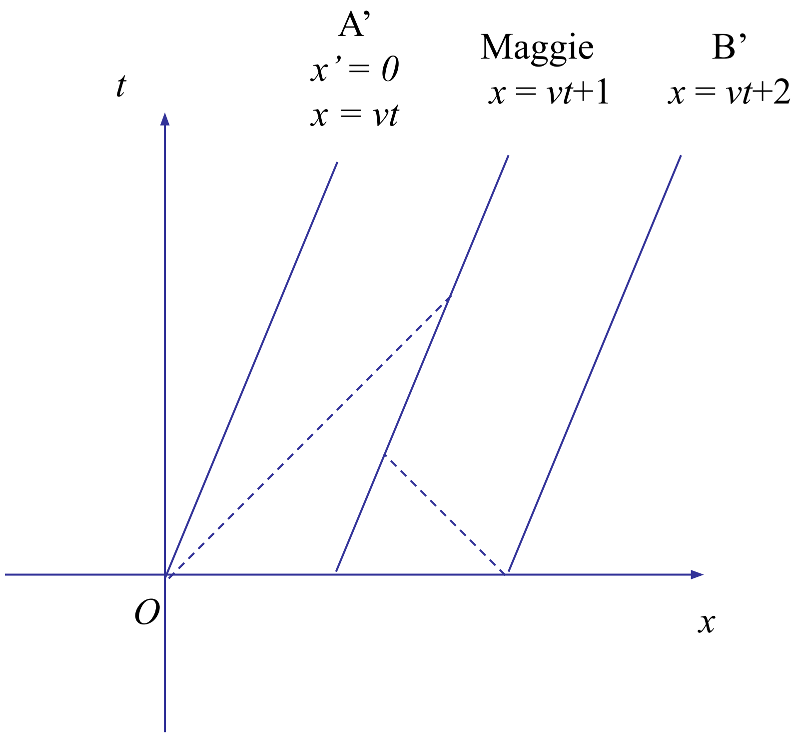
Suppose at noon, Maggie happens to reach point C. The light ray coming from the left will reach Maggie a little later than the light ray coming from the right (the speed of light is fixed).

Therefore, Maggie will conclude that the two clocks are not synchronized, because the two light flashes reach her at two different times.



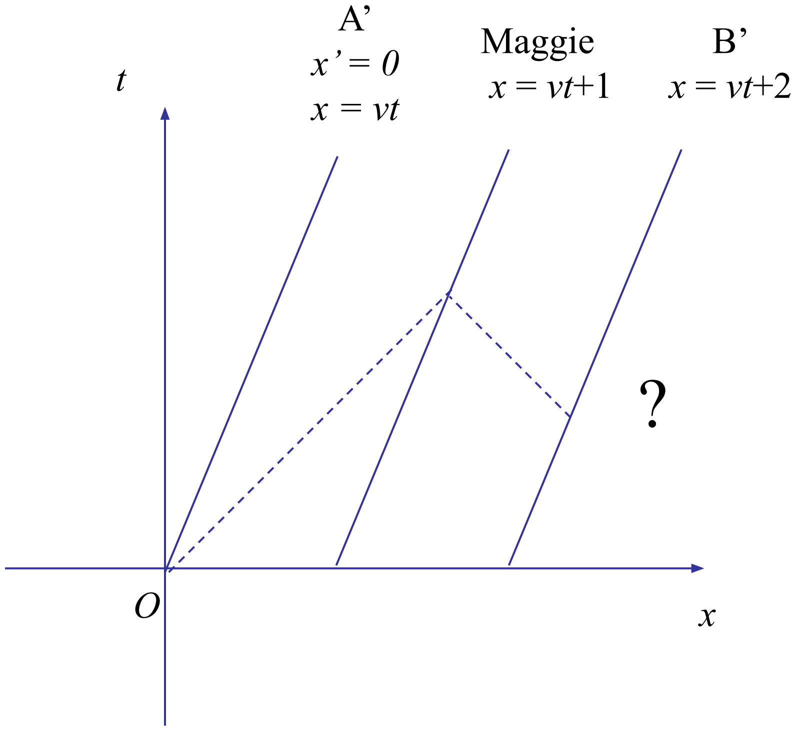
Space-time diagram





How to make the two clocks sync to Maggie?

Suppose we have the freedom to choose when and where B' release the light, what choice should we make such that the two clocks are synchronized to Maggie?

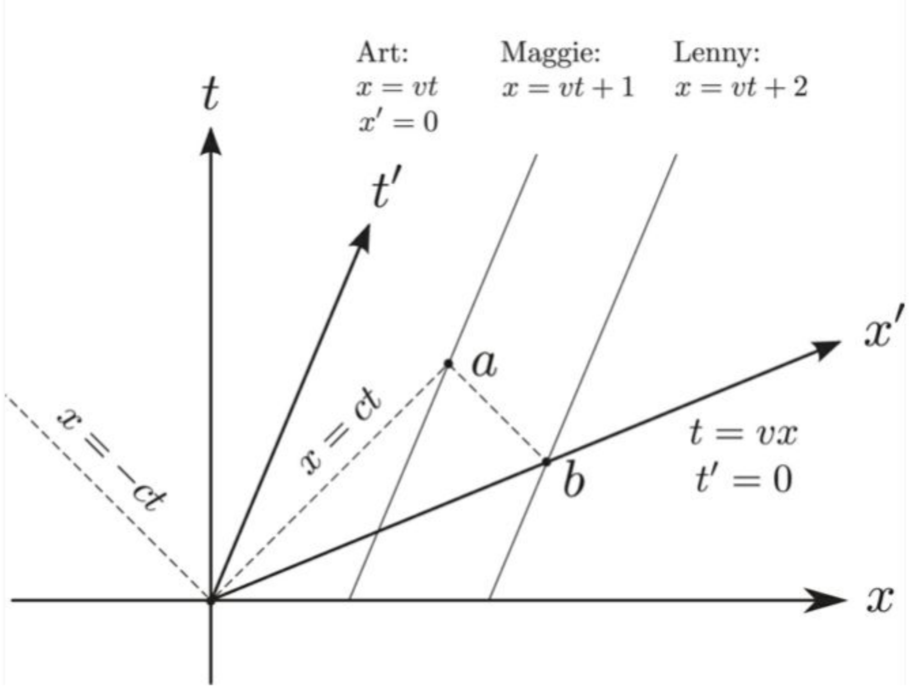


Derivation

What is the x’ axis?

For the x-t frame, all the clocks on the $t = 0$ (x axis) are synchronized.

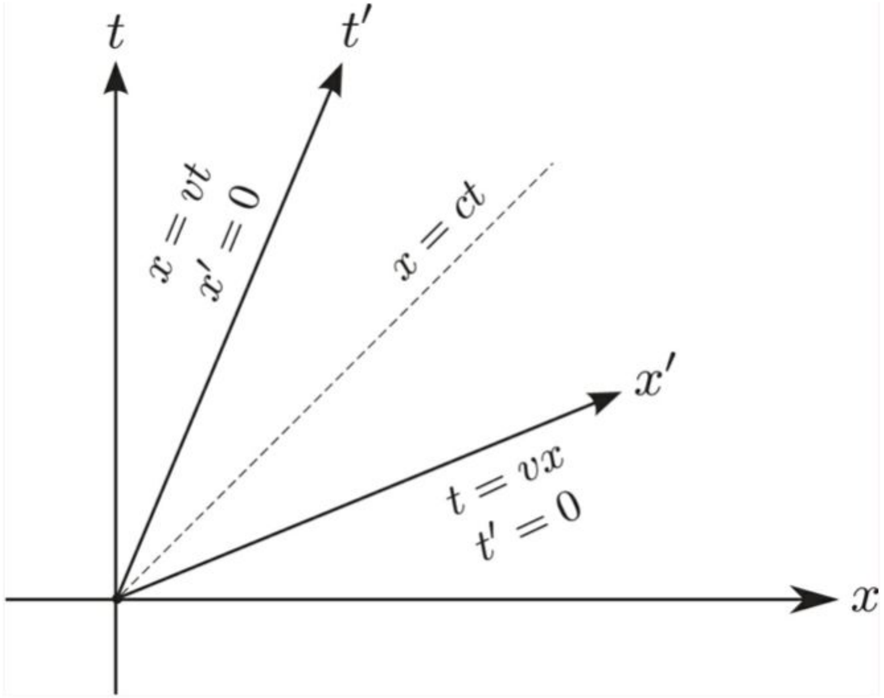
For the x'-t' frame, all the clocks on the $t = vx$ are synchronized. This is the $t' = 0$ (x' axis).



What is the x' axis?

In the ground frame, *x axis and t axis have equal angles to* the world line of light x=t.

In the moving frame, t' axis (x'=0) is x=vt, and x' axis (t'=0) is $t = vx$. Again, they have equal angle to the world line of light x=t.



The speed of light is unchanged under any frame (always 45 degree).

The synchronization depends on frame choice.

Lorentz transformation

How to transform from x-t frame to x'-t' frame?

Recall the Newtonian equation, $x' = x - vt$. At $x' = 0$, we have $x = vt$. Our new relation also needs to satisfy this condition. Therefore,

$$x' = (x - vt)\, f(v)$$

where $f(v)$ is any function. Next, Einstein realized nothing in physics distinguishes positive and negative velocity. Therefore $f(v)$ should be a function of v^2.

$$x' = (x - vt)\, f(v^2)$$

Lorentz transformation

What about t'? We know at $t'=0$, $t=vx$. Therefore

$$t' = (t-vx)\, g(v^2).$$

Next we demand a light ray that satisfies $x=t$ must also satisfy $x'=t'$. Therefore $f(v^2)=g(v^2)$.

To sum up we have $x' = (x-vt)\, f(v^2)$; $t' = (t-vx)\, f(v^2)$.

Lorentz transformation

Next Einstein realized my frame is moving relative to you with velocity v is the same as your frame is moving relative to me with velocity $-v$.

From $x' = (x-vt)\, f(v^2)$; $t' = (t-vx)\, f(v^2)$, we get the following inverse transformation.

$x=(x'+vt')\, f(v^2)$; $t = (t'+vx')\, f(v^2)$

Finally we plug in x', t' into x expression.

$x=[(x-vt)\, f(v^2)+v(t-vx)\, f(v^2)]\, f(v^2)$ => $x = xf^2(v^2) - v^2xf^2(v^2) - vtf^2(v^2) + vtf^2(v^2)$ =>
$f(v^2)=(1-v^2)^{-1/2}$

Therefore

$$x' = \frac{x - vt}{\sqrt{1 - v^2}} \qquad t' = \frac{t - vx}{\sqrt{1 - v^2}}$$

$$x = \frac{x' + vt'}{\sqrt{1 - v^2}} \qquad t = \frac{t' + vx'}{\sqrt{1 - v^2}}$$

$$(y' = y, z' = z)$$

Velocities

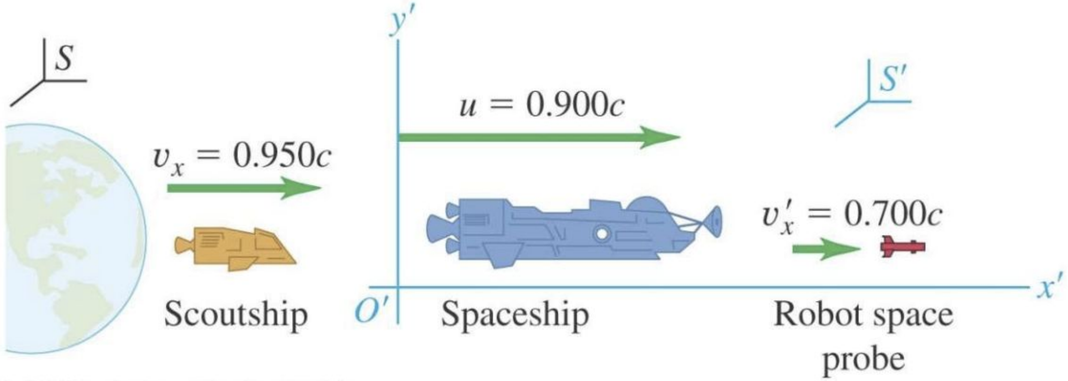
Taking the time derivatives of the Lorentz transformation, we have

$$v'_x = \frac{v_x - v}{1 - v v_x} \qquad v_x = \frac{v'_x + v}{1 + v v'_x}$$

$$v'_y = \frac{v_y\sqrt{1 - v^2}}{1 - v v_x} \qquad v_y = \frac{v'_y\sqrt{1 - v^2}}{1 + v v'_x}$$

$$v'_z = \frac{v_z\sqrt{1 - v^2}}{1 - v v_x} \qquad v_z = \frac{v'_z\sqrt{1 - v^2}}{1 + v v'_x}$$

Example



a) What is the probe's velocity relative to earth?
b) What is the scoutship's velocity relative to spaceship?

Length contraction

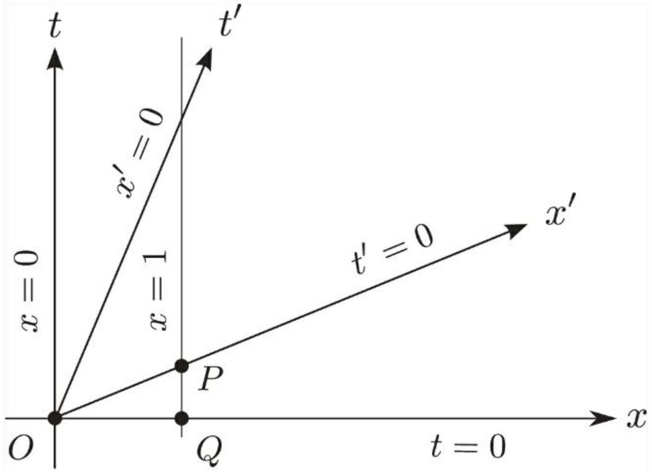


Figure 1.5: Length Contraction.

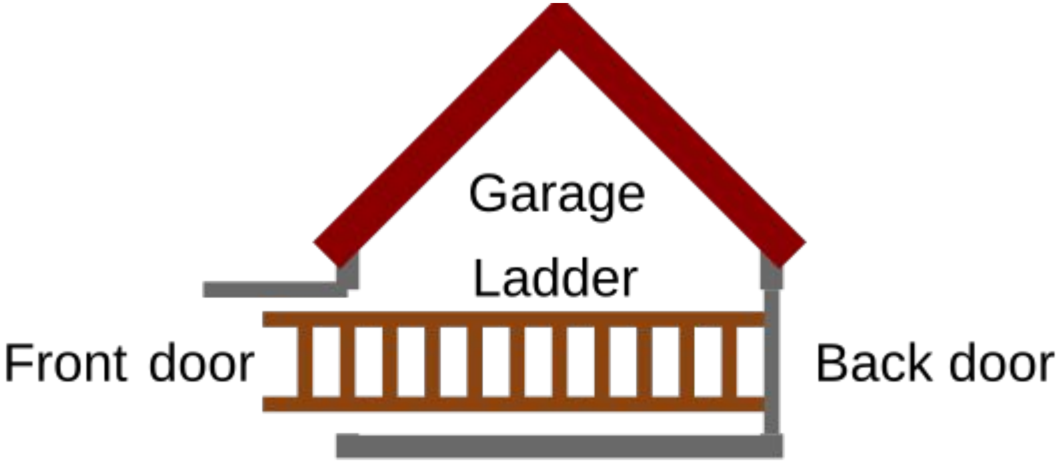
$t' = 0$ means $t = vx$

$$x' = \frac{x - v^2 x}{\sqrt{1 - v^2}} = \sqrt{1 - v^2}\, x$$

Ladder paradox

A ladder, parallel to the ground, travelling horizontally at relativistic speed (near the speed of light) and therefore undergoing a Lorentz length contraction.

The ladder is imagined passing through the open front and rear doors of a garage or barn which is shorter than its rest length, so if the ladder was not moving it would not be able to fit inside.

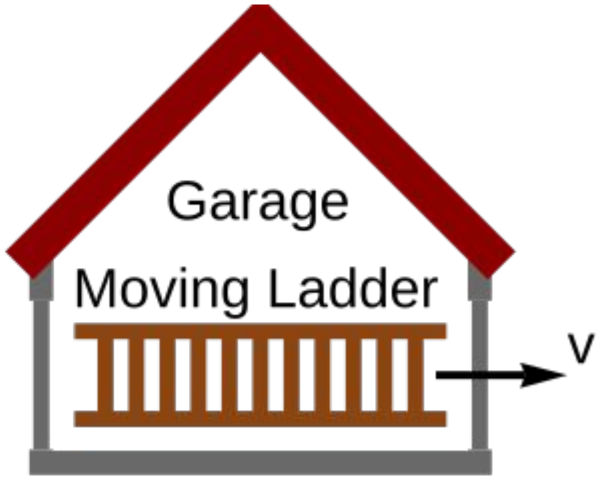


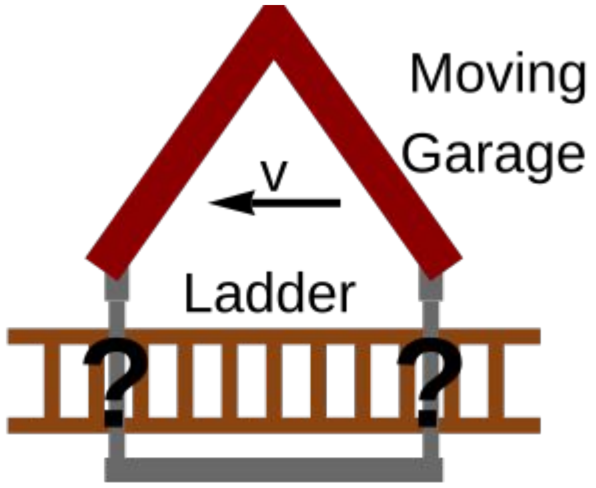
Ladder paradox

To a stationary observer, due to the contraction, the moving ladder is able to fit entirely inside the building as it passes through.

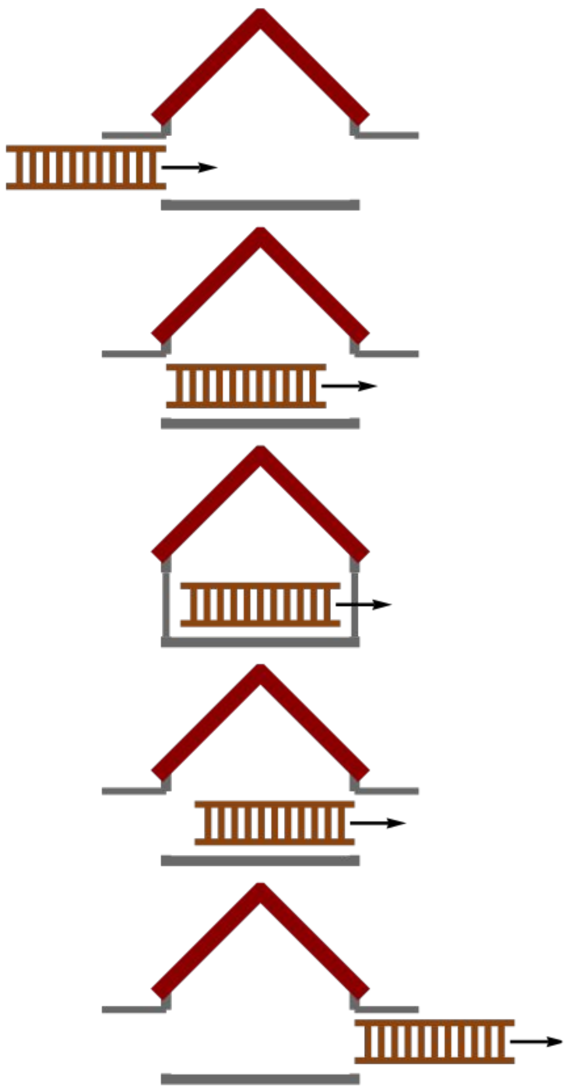
On the other hand, from the point of view of an observer moving with the ladder, the ladder will not be contracted, and it is the building which will be Lorentz contracted to an even smaller length. Therefore, the ladder will not be able to fit inside the building as it passes through.

This poses an apparent discrepancy between the realities of both observers.

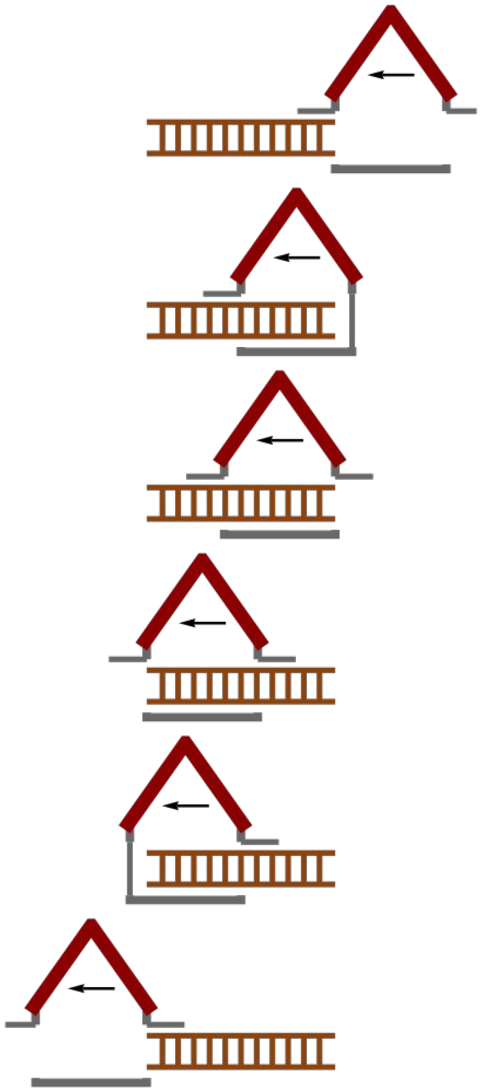




Resolution



Scenario in the garage frame: a length contracted ladder passing through the garage



Scenario in the ladder frame: a length contracted garage passing over the ladder. Only one door is closed at any time

Time dilation

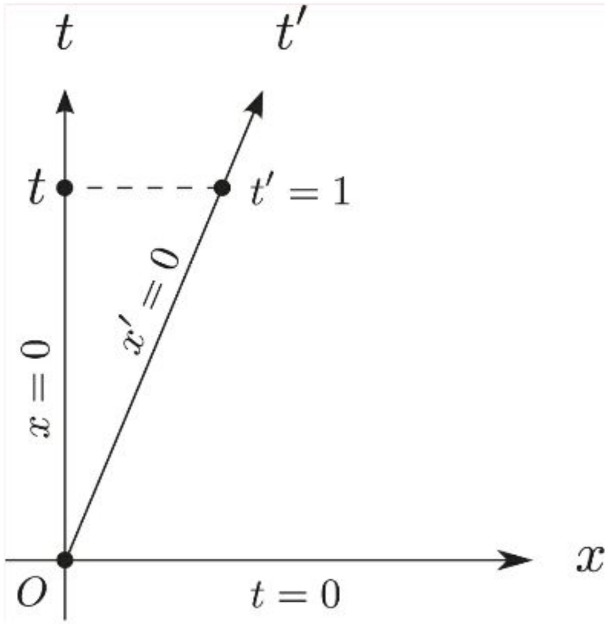


Figure 1.7: Time Dilation.

Given $x' = 0$

$$t = \frac{t'}{\sqrt{1 - v^2}}$$

Twin paradox

Two twins are at rest in an inertial reference frame. Both carry physical clocks that are synchronized. One of the twins stays put, we will call this twin A. The other twin, who we will call B, takes a trip and goes away for a while and then returns, usually at relativistic speeds to bring out the paradox.

When the twins are back together, they compare their clocks and they find that the clock of B shows less time to have elapsed than the clock of A, thus B is younger.

However, all motions are relative. In B’s frame, A is traveling with relativistic speeds, thus A should be younger upon return.

Resolution

For the twins to meet again, they must experience accelerations. This means the twins are not always in inertial frames.

Accelerations are beyond the coverage of special relativity. It is a topic covered by general relativity.

Say B turns around because of a massive star. The star is static to A but is not static to B. That breaks the symmetry of relative motion between A and B.

Under general relativity, the elapsed time of B is smaller than that of A.

Summary

Invariance of physical laws

Simultaneity

Lorentz transformation

Lorentz velocity transformation

Length contraction

Time dilation